Wagon-wheel effect



https://en.wikipedia.org/wiki/Wagon-wheel_effect

Temporal aliasing – helicopter blades



https://www.youtube.com/watch?v=yr3ngmRuGUc

Aliasing in practice



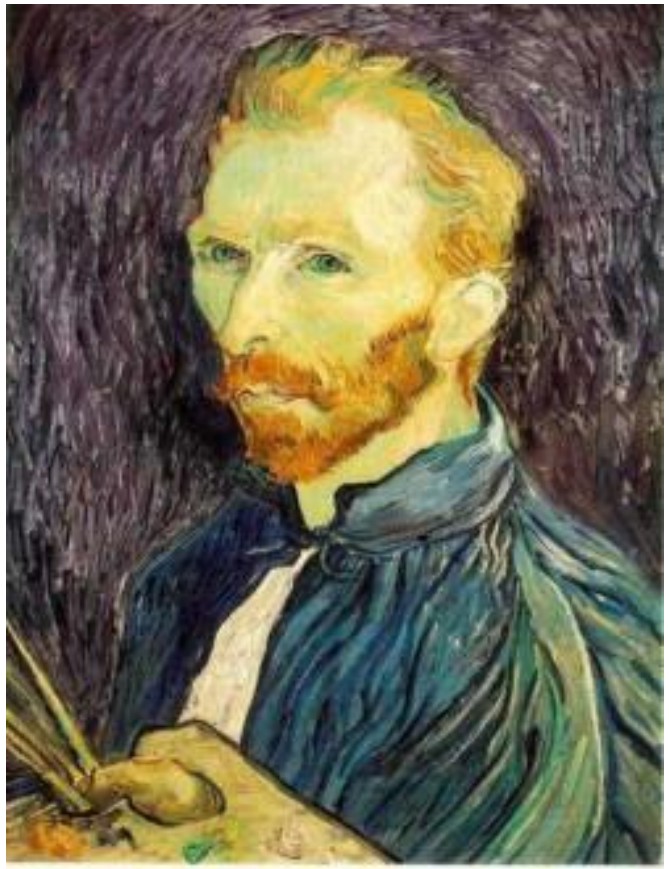


Aliasing

- When downsampling by a factor of two
  - Original image has frequencies that are too high

- How can we fix this?

Gaussian pre-filtering



Gaussian 1/2



G 1/4

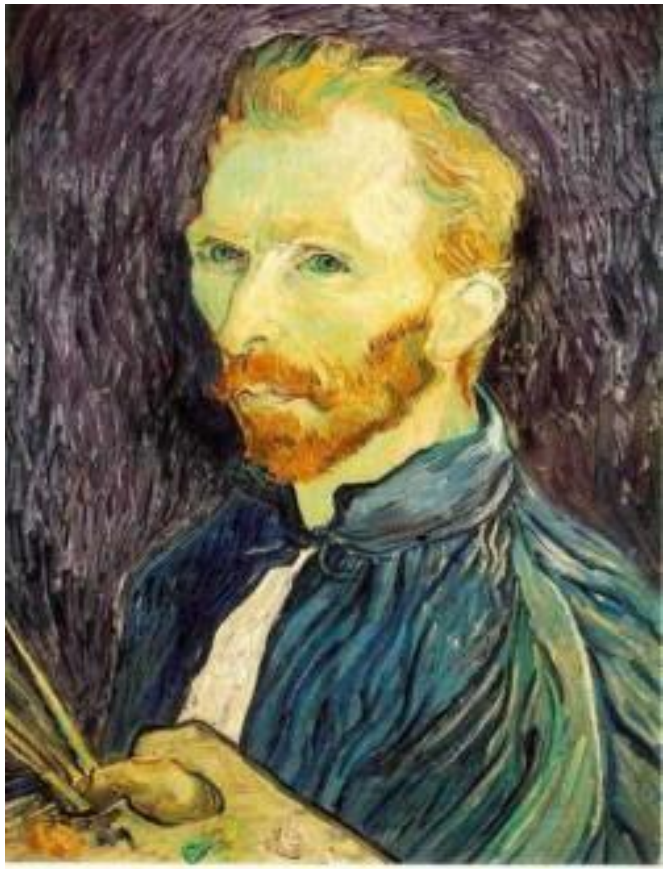


G 1/8

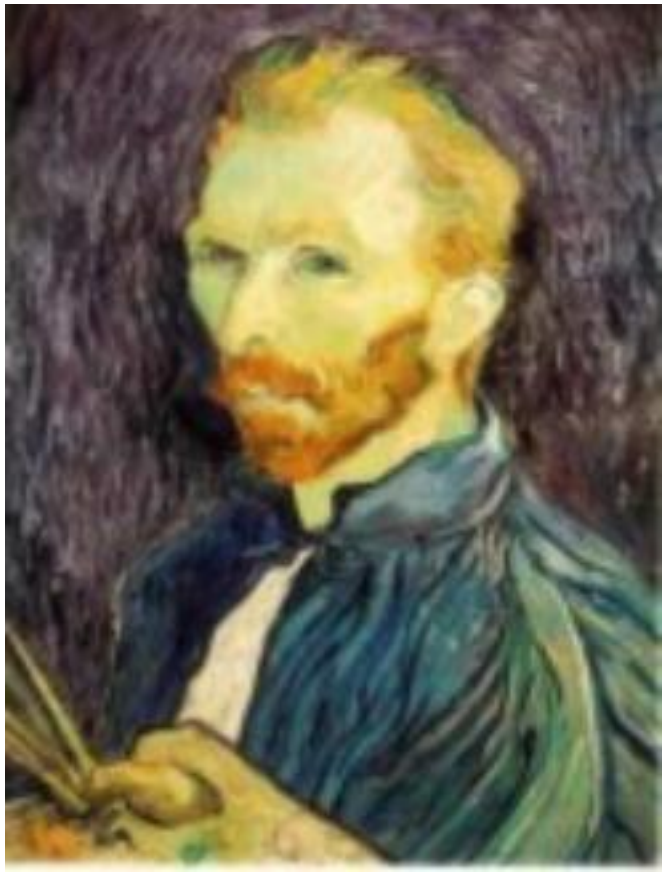
- Solution: filter the image, *then* subsample

Source: S. Seitz

Subsampling with Gaussian pre-filtering



Gaussian 1/2



G 1/4

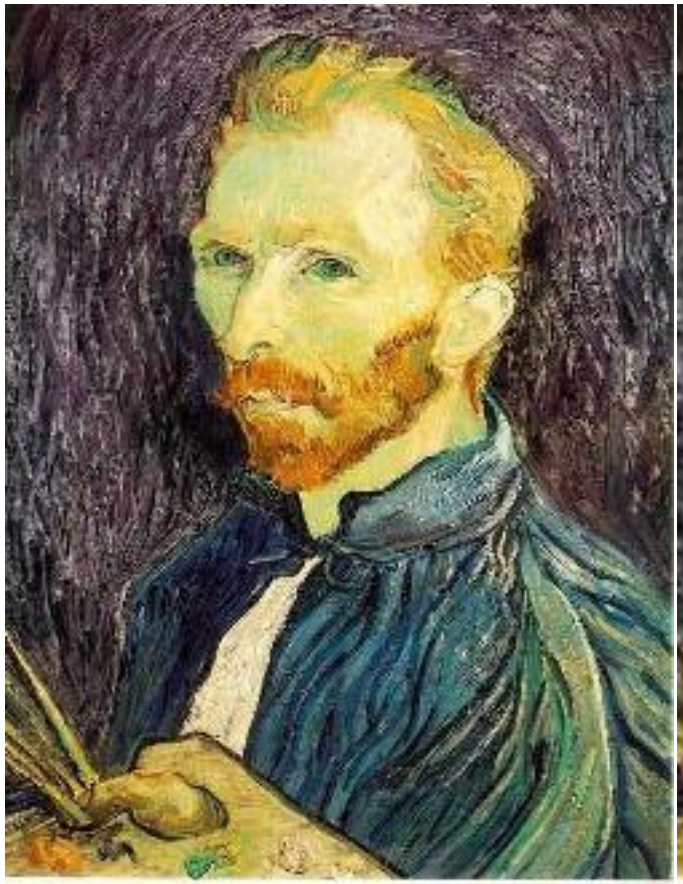


G 1/8

- Solution: filter the image, *then* subsample

Source: S. Seitz

Compare with...



1/2



1/4 (2x zoom)



1/8 (4x zoom)

Source: S. Seitz

Gaussian pre-filtering

- Solution: filter the image, *then* subsample

